
Q/R/M/C: A Stage-Decomposition Evaluation Protocol for Memory-Based Agents

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We present Q/R/M/C, an evaluation protocol for memory-based agents that sep-
2 arates task completion into four observable stages: Query formation, Retrieval
3 satisfaction, target Materialization, and Completion. The protocol is motivated
4 by a practical limitation of end-to-end success: the same failure score may arise
5 because an agent did not formulate a usable query, because its memory returned no
6 relevant evidence, because the planner committed to the wrong target, or because
7 execution failed after a correct commitment. When a system exposes an explicit
8 query–lock chain, Q/R/M/C represents these stages as nested episode events and es-
9 timates their conditional transition rates from ordinary trajectory logs. We validate
10 the resulting diagnoses through targeted interventions rather than by descriptive
11 correlation alone. In MiniGrid and BabyAI, oracle-stage interventions distinguish
12 retrieval-limited from downstream-limited tasks; in the dominant retrieval failure,
13 candidate ranking is 96.9% accurate when candidates exist, whereas retrieval re-
14 turns an empty candidate set at 91% of decision points. In a 35,640-run multi-agent
15 study, faster peer sharing raises $P(M^* | R^*)$ but lowers $P(C^* | M^*)$, revealing a
16 shift from incomplete commitment to contention after commitment. Provenance
17 stratification locates the loss in peer-supported target locks, and a reservation-and-
18 rollback ablation provides intervention-based evidence that occupancy contention,
19 rather than evidence quality, is the governing mechanism. Finally, the same event
20 roles are instantiated on OpenAI SDK, LangGraph, and AutoGen tool traces. Struc-
21 tural retrieval and execution faults produce the predicted profiles across all three
22 stacks. On the 63 scored cells of a blinded fault benchmark, Q/R/M/C reaches
23 0.873 exact-stage accuracy and 0.835 macro-F1, exceeding progress, two-stage,
24 conformance, and learned diagnostic baselines, and its diagnoses select repairs
25 with a mean completion gain of 0.293 and regret 0.061 relative to the best available
26 repair arm. Systems without an explicit lock chain are reported through marginal
27 event frequencies, which are kept separate from the conditional-rate product.

28 1 Introduction

29 A central difficulty in evaluating memory-based agents is that task success provides little information
30 about how memory affected the decision process. An agent may fail before retrieval because its
31 current intent does not produce a usable query; it may query correctly but retrieve no relevant record;
32 it may retrieve appropriate evidence yet commit to the wrong target; or it may select the correct
33 target and still fail during execution. These outcomes are qualitatively different, but conventional
34 end-to-end evaluation maps all of them to the same scalar failure.

35 This limitation becomes more consequential when memory is distributed across several agents.
36 Sharing newly discovered evidence can improve target selection, yet the same shared evidence may

37 cause several agents to commit to the same scarce target. A single success rate then conflates the
38 benefit of information consolidation with the cost of coordination. The resulting ambiguity is not
39 only descriptive: it makes it difficult to decide whether the next system modification should improve
40 memory accumulation, retrieval, commitment, local control, or coordination after commitment.

41 We address this problem by partitioning memory-based task completion into four interaction events
42 observable at the instrumented agent–memory–environment interfaces, rather than into hidden
43 computational phases. Query formation (Q) records whether the agent issues a semantic intent that
44 can be evaluated against memory. Retrieval satisfaction (R) records whether the returned candidate
45 set contains evidence satisfying the task. Target Materialization (M) records whether the planner
46 converts that evidence into a concrete commitment. Completion (C) records whether the committed
47 target is reached. For systems with an explicit query–lock chain, the corresponding episode events are
48 nested and define a conditional decomposition of semantic-path completion. The partition is semantic
49 rather than architectural: each event marks a distinct commitment in the interaction among the agent,
50 its memory, and the environment, and none of the definitions refers to internal computational state.
51 The arithmetic is the ordinary chain rule; the methodological contribution lies in specifying the events,
52 defining when the rates are comparable, and validating their interpretation through interventions.

53 The study follows the same progression at three levels. First, we test whether stage profiles correctly
54 identify controlled failures in a single symbolic agent. Second, we use the protocol to study how
55 communication cadence changes the location of the bottleneck in a multi-agent collective and whether
56 the inferred mechanism survives a targeted coordination intervention. Third, we examine whether
57 the event roles remain usable when the underlying runtime is replaced by off-the-shelf tool-based
58 agent frameworks. The experiments are therefore organized around diagnostic validity, mechanism
59 identification, and instrumentation portability rather than around a new navigation architecture.

60 2 Related Work

61 Explicit memory has been studied in cognitive and semantic mapping, goal-conditioned reinforcement
62 learning, embodied navigation, multi-agent communication, and LLM-agent systems. Cognitive-map
63 and semantic-map methods motivate place abstraction and retrieval by meaning [Tolman, 1948, Gupta
64 et al., 2017, Savinov et al., 2018]; goal-conditioned reinforcement learning conditions behavior on
65 target descriptions or embeddings [Andrychowicz et al., 2017]; multi-agent reinforcement learning
66 studies both learned communication channels [Sukhbaatar et al., 2016, Foerster et al., 2018] and
67 distributed exchange protocols [Boyd et al., 2006]; and LLM agents increasingly rely on explicit
68 memory layers while remaining evaluated mainly by end-to-end task outcomes [Wang et al., 2024,
69 Shinn et al., 2023, Packer et al., 2023, Park et al., 2023, Wu et al., 2023]. These lines primarily design
70 memory or communication mechanisms. Our work instead evaluates where a given mechanism fails
71 once a task requires memory retrieval and target commitment.

72 The closest evaluation literature decomposes retrieval-augmented generation and multi-step agent
73 behavior. RAGAS separates retrieval and generation quality [Es et al., 2024], RAGChecker introduces
74 claim-level diagnostics evaluated against human judgments [Ru et al., 2024], and AgentBoard replaces
75 final success with analytic progress rates for multi-turn LLM agents [Ma et al., 2024]. Q/R/M/C
76 differs in that its variables are logged actions of a task-executing agent, not LLM-based text-quality
77 judgments; it separates commitment from retrieval; and it explicitly represents the two multi-agent
78 channels that matter after retrieval, namely shared evidence and target occupancy. The four stages are
79 therefore closer to a memory-and-control interface than to a generic progress score.

80 Process mining and conformance checking also compare event logs with structured stage models
81 [van der Aalst, 2016, Carmona et al., 2018], while process reward models score intermediate reasoning
82 steps for training [Lightman et al., 2024]. Q/R/M/C adopts the log-level discipline of process analysis
83 but uses agent-specific event semantics and evaluates the stages at test time. The same stance —
84 specifying and checking externally observable events rather than internal state — is standard in
85 runtime verification and distributed-system observability [Leucker and Schallhart, 2009]; Q/R/M/C
86 applies it to the memory interface of task-executing agents. Its diagnoses are checked through
87 oracle substitutions, provenance analysis, and mechanism-targeted interventions. This distinction
88 is important because a stage profile is useful only if changing the identified stage changes behavior
89 in the predicted direction. A recent intervention-based line makes a related move at a different
90 granularity: REFLECT attributes an error to a candidate step in an LLM agent trace and verifies

the attribution by replaying the trace with a diagnosis-specific patch [Lin et al., 2026]. $Q^*/M^*/C^*$ measures recurring pipeline stages rather than individual steps and computes its rates from logs alone, reserving interventions for validating the protocol and probing mechanisms rather than for producing each diagnosis.

3 Contributions and Empirical Claims

We make three empirical and methodological claims.

Claim 1: observable stage decomposition. For agents that expose an explicit query–retrieve–commit–execute chain, semantic-path completion can be represented by nested events Q^* , R^* , M^* , and C^* . Their conditional frequencies are directly estimable from logs, while a conditional stage-Markov assumption supports the stronger interpretation of each link as a stable property of the subsystem that owns the transition. The event contract makes a practically useful stage decomposition measurable and interventionally testable. The events are defined over interaction observable at instrumented interfaces, not over internal computational phases, so the decomposition claims nothing about how the agent deliberates.

Claim 2: communication moves the bottleneck. In a collective with scarce targets, faster peer broadcast improves commitment from retrieved evidence but reduces completion after commitment. Formally, as the broadcast interval K decreases, $P(M^* | R^*)$ increases and $P(C^* | M^*)$ decreases. The claim concerns the two conditional links, not their product. We further claim that the completion loss is governed by occupancy contention around peer-supported commitments; this mechanism is evaluated by retaining peer evidence while changing the coordination rule.

Claim 3: event roles are portable, numerical regimes are not interchangeable. The semantic roles of query, satisfied retrieval, commitment, and completion can be instantiated on independent tool-based agent frameworks through system-specific adapters. When a runtime does not expose a nested lock chain, the same roles are summarized by marginal event frequencies rather than by the conditional product. Portability therefore means a shared diagnostic vocabulary and adapter contract, not unrestricted numerical comparison across incompatible logging regimes.

The primary evidence is obtained in instrumented grid-world substrates, where entity correspondence and stage events are observable. Continuous-control, LLM-collective, and third-party framework experiments are used as portability and robustness checks. The limitations of this scope are discussed together with the results and expanded in the supplement.

4 Method and Materials

4.1 Evaluation protocol

A memory-based episode consists of four operations. The agent first issues a semantic query, represented in our symbolic systems by required, preferred, and penalty tags. Memory then returns a candidate set. The planner commits to one candidate by materializing it as an actionable target, and the controller executes until the target is reached or the episode terminates. The corresponding episode events are defined as follows: Q^* is recorded when the intent is issued; R^* occurs when the returned set contains a task-satisfying candidate; M^* occurs when the planner locks a true target drawn from that set; and C^* occurs when the controller reaches the locked target. An episode may contain several query–retrieve–lock attempts. A stage is counted only when one attempt realizes it together with its predecessors, so

$$C^* \subseteq M^* \subseteq R^* \subseteq Q^*. \quad (1)$$

The final event is semantic-path completion rather than raw task success, because an episode may succeed without using the explicit retrieval path.

When the chain is observable, we estimate

$$\begin{aligned} q &= P(Q^*), & r &= P(R^* | Q^*), \\ m &= P(M^* | R^*), & c &= P(C^* | M^*). \end{aligned} \quad (2)$$

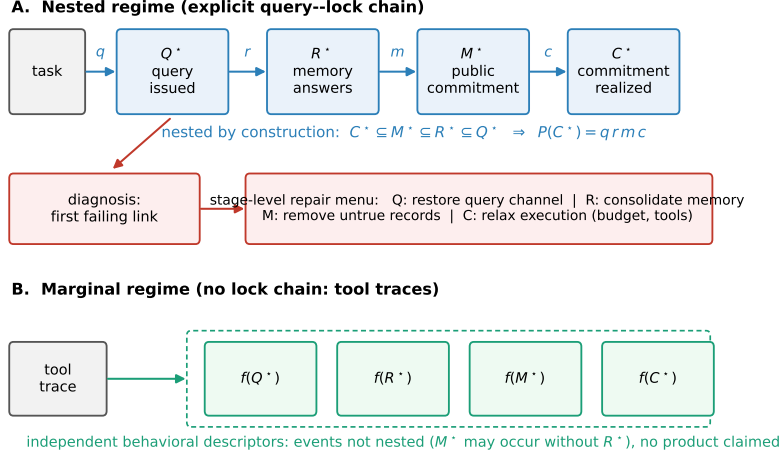


Figure 1: The Q/R/M/C protocol. (A) For an observable query–lock chain, nested events define conditional link rates whose product is semantic-path completion, and a first-failing-link rule yields a stage diagnosis. (B) For tool traces without nesting, the same roles are reported only as marginal frequencies.

135 The nesting gives the exact identity

$$P(C^*) = q r m c. \quad (3)$$

136 No stage-Markov assumption is required for this identity. The assumption is used only when a link
 137 rate is interpreted as a stable property of a subsystem and used to predict a targeted intervention. A
 138 simulation stress test in the supplement introduces a hidden confounder and measures the resulting
 139 bias; under that violation, m remains stable while c is inflated, which identifies the M–C contrast as
 140 the first interpretation to become unreliable.

141 The four events admit a uniform reading as semantically distinct acts in the observable interaction.
 142 Q^* records an informational commitment: a query stating what evidence is needed. R^* records the
 143 memory’s answer to that commitment. M^* records a public commitment to act on specific evidence;
 144 the commitment exists from the moment the lock is written, and in a collective it becomes visible
 145 to peers and occupies a target. C^* records the environment’s confirmation that the commitment
 146 was realized. This reading grounds Materialization independently of any cognitive interpretation:
 147 the moment at which an agent “decides” is not observable, whereas the appearance of a public
 148 commitment is a log event. Such commitments are also the source of the occupancy contention
 149 studied in the multi-agent experiments.

150 Some tool-based agents expose the four semantic roles but not a nested lock chain. In this case we
 151 report marginal frequencies $f(Q^*)$, $f(R^*)$, $f(M^*)$, and $f(C^*)$. An agent may, for example, guess
 152 the correct destination without retrieving supporting evidence, so M^* can occur when $R^* = 0$. These
 153 marginals are treated as a behavioral stage profile and are never substituted into the conditional-rate
 154 product.

155 For N communicating agents, the same events are logged per agent. The conditioning context is
 156 enlarged by two channels: peer memory $\mathcal{M}_{-i}^{(K)}$ received under broadcast interval K , and target
 157 occupancy $\mathcal{O}_{-i}$ induced by the commitments of the other agents. This separation is central to the
 158 multi-agent experiment: memory sharing can improve the R–M transition, whereas occupancy affects
 159 the M–C transition.

160 4.2 Single-agent materials and interventions

161 The single-agent benchmark contains four semantic task families. Water and rest require retrieval of
 162 resource-like locations; goal-region rejoin in FourRooms requires returning to a semantically defined
 163 region; and hazard recovery in LavaGapS7 requires retrieving and materializing a post-hazard route.

Each task is evaluated with 10 seeds and 8 episodes per seed. A BabyAI implementation is used as a portability check in the supplementary material.

We apply stage-specific oracle interventions to the two most difficult families. Oracle-R replaces the retrieved set with a correct task-satisfying candidate, oracle-M replaces the planner commitment with a correct target lock, and oracle-C replaces post-lock execution with direct arrival. Each intervention preserves the remaining interfaces, allowing the change in semantic-path completion and end-to-end success to be attributed to one stage. A behavior-cloned policy is included as an observability control: it can reach competitive success without exposing an explicit query or commitment interface.

4.3 Multi-agent materials

The multi-agent task is a scarce semantic water-search problem on one symbolic substrate. We compare four memory organizations: independent private memories, a shared event bus, a central trust-weighted aggregator, and peer-to-peer snapshot exchange. The peer architecture is evaluated at broadcast intervals $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$. The main design crosses $N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts, three hazard densities, and 40 seeds, producing 35,640 runs. A scale-up to $N = 12$ adds 8,640 runs and is reported in the supplement.

To study the source of a commitment, every M^* event is annotated with the foreign share φ of its supporting evidence mass. The share is computed from the merge weights used by the memory system and is frozen when the target is locked. We call locks with $\varphi \geq 0.8$ *warps*. The threshold-sensitivity analysis covers $\varphi \in \{0.5, \dots, 0.95\}$. A decentralized reservation-and-rollback mechanism is then used as a mechanism-targeted intervention: a lock carries a reservation on the next scheduled broadcast, and a later lock dominated by an earlier reservation is retracted with exponential backoff. The intervention retains peer evidence while changing target coordination.

We additionally evaluate a CommNet-PPO policy on the same scenario and a three-agent llama3.1 collective on a text substrate with and without shared symbolic memory. These experiments are not performance claims against learned or LLM systems; they test whether the diagnostic distinction between observable stages, collapsed stages, and coordination after commitment remains meaningful beyond the primary symbolic implementation.

4.4 Third-party agent stacks

The third-party study uses one semantic event contract and a system-specific trace adapter. The invariant roles are: a memory query is issued; returned evidence correctly identifies the task-relevant place; the agent commits to that place; and the task is completed. The same adapter instruments an OpenAI SDK function-calling loop, a LangGraph ReAct agent, and AutoGen. Each runtime accesses an episodic household memory through `recall`, `goto`, and `take`. A query records Q^* ; a returned record naming the true place records R^* ; `goto` to the true place records M^* ; and a successful `take` within the tool budget records C^* .

Four controlled variants are used: a clean control, a consolidation gap that removes the needed memory fact, a stale ambiguity that introduces an outdated distractor, and a tight tool budget that prevents completion. Each cell contains 20 episodes. The complete 240-episode experiment is run with llama3.1 and repeated with qwen3:4b. In the blinded study, ten structural and behavioral fault types are injected across the four stages. A first-failing-stage rule and all thresholds are fixed before execution. The rule receives only anonymized marginal profiles and one labeled control cell; a second control is included in the blind set to measure false positives. The design yields $11 \times 3 \times 2 = 66$ cells; three are excluded by a manipulation check fixed before scoring — the prompt-only fault does not change llama3.1’s behavior (its query rate drops by less than 0.20 against the calibration control) — leaving 63 scored cells. The exclusion applies to every diagnostic method alike.

4.5 Diagnostic baselines and repair selection

Four stage-localization baselines and a success-only reference are evaluated on the same blind set. Each rule uses a pre-specified mapping and receives the same calibration control; no threshold is tuned on the scored fault cells. The baselines are a success-only fault detector, an AgentBoard-inspired first-drop milestone rule, a RAG-inspired two-side rule over retrieval (Q, R) versus execution (M, C),

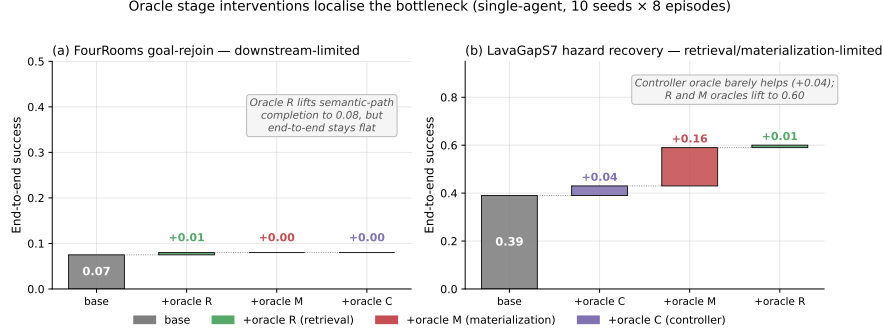


Figure 2: Oracle-stage interventions. Hazard recovery is retrieval-limited: oracle-R raises end-to-end success from 0.39 to 0.60. Goal-region rejoin remains downstream-limited.

214 a conformance rule for the first deviation from the expected event order, and a multinomial logistic
 215 regression trained leave-one-fault-type-out on per-cell event features.

216 Repair utility is measured by mapping each diagnosis to a fixed stage-level repair: restore the query
 217 channel (Q), consolidate memory with reliable recall (R), remove non-true-place records (M), or
 218 relax execution with a larger budget and reliable tools (C). Every repair arm is run for every llama3.1
 219 condition on all three stacks (2,640 repair episodes). We report mean completion gain over no
 220 repair and regret relative to the best arm in the same fault cell, with paired bootstrap over the 27
 221 fault-framework cells.

222 5 Results and Discussion

223 5.1 Single-agent stage attribution

224 End-to-end success differs across task families but does not explain the difference. Water and rest
 225 reach success rates of 0.95 and 0.93, whereas goal-region rejoin and hazard recovery are substantially
 226 weaker. Oracle interventions separate the latter two failures. In hazard recovery, oracle retrieval raises
 227 success from 0.39 to 0.60, with bootstrap 95% confidence intervals [0.35, 0.43] and [0.51, 0.69]; the
 228 paired change is positive in 9/10 seeds and the exact sign test gives $p = 0.004$. Oracle-C produces
 229 only a small improvement, placing the primary failure before local execution. In goal-region rejoin,
 230 oracle retrieval restores semantic-path completion without changing end-to-end success, which places
 231 the remaining failure after retrieval rather than in query formation.

232 The retrieval diagnosis can be refined using the same logs. When a candidate set is non-empty,
 233 the selected candidate is correct in 96.9% of cases. However, the set is empty at 91% of decision
 234 points. The estimate is stable when calls are pooled (0.908), seeds are weighted equally (0.909), or
 235 episodes are weighted equally (0.891), which rules out the explanation that long failing episodes
 236 dominate the count. The binding R-stage failure is therefore evidence accumulation into a queryable
 237 memory structure, not candidate ranking. This result also illustrates the intended resolution of the
 238 protocol: Q/R/M/C identifies the failed link, while a within-link trace analysis distinguishes candidate
 239 generation from ranking.

240 The behavior-cloning baseline reaches competitive task success but does not expose explicit query
 241 and commitment events. Q/R/M/C therefore describes its limited stage observability rather than
 242 forcing an artificial four-link decomposition onto the policy. This is a boundary of the protocol, not a
 243 performance disadvantage of the learned baseline.

244 5.2 Communication cadence and bottleneck migration

245 The 35,640-run sweep produces a consistent stage-level separation between fast and slow sharing.
 246 Retrieval is nearly saturated across architectures, while the largest differences occur at Materialization
 247 and Completion. Across peer runs, the Spearman association between $P(M^* | R^*)$ and K is
 248 -0.53 , with cluster-robust 95% confidence interval $[-0.59, -0.47]$ over 81 configuration cells. The
 249 corresponding association for $P(C^* | M^*)$ is $+0.43$, with interval $[+0.37, +0.50]$. Thus, faster

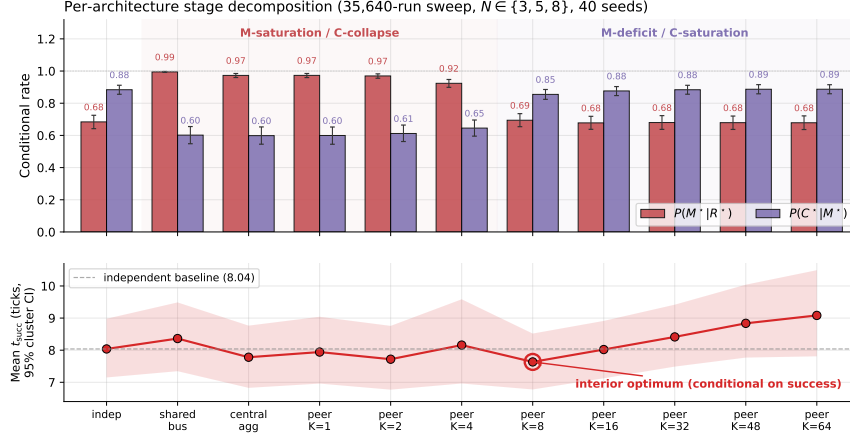


Figure 3: Conditional Materialization and Completion rates by memory architecture, with mean time-to-success among successful runs. Fast sharing saturates M but collapses C; $K = 8$ minimizes conditional time-to-success, while a censoring-aware metric changes the ordering.

250 sharing improves commitment from retrieved evidence but reduces completion after commitment, as
 251 predicted by Claim 2.

252 The relation is also visible within fixed-cadence groups. The Pearson correlation between the M and
 253 C links reaches -0.61 at $K = 4$ and approaches noise by $K = 64$. Increasing the population to
 254 $N = 12$ sharpens the maximum negative relation to -0.70 and shifts it to $K = 8$. These observations
 255 support a migration of the lower conditional link rather than a uniform communication benefit.

256 The practical optimum depends on the metric. Among successful runs, mean time-to-success is
 257 minimized at $K = 8$ (7.63 ticks versus 8.04 for independent memory). Cell-paired contrasts give
 258 -0.39 ticks relative to $K = 16$, with 95% confidence interval $[-0.80, -0.08]$, and -0.40 relative to
 259 independent memory, with interval $[-0.59, -0.24]$; $K = 8$ is statistically tied with $K \in \{2, 4\}$. This
 260 metric conditions on success. When unsuccessful runs are assigned the full step limit, independent
 261 memory is marginally better and $\Delta(K = 8 - \text{independent}) = +0.32$ with interval $[+0.05, +0.63]$.
 262 The interior optimum is therefore a result about conditional speed, not a universal optimum of
 263 communication cadence.

264 5.3 From provenance association to coordination mechanism

265 Provenance stratification identifies the subset of commitments in which the Completion loss is
 266 concentrated. The warp share is 0.55 at $K = 1$ and 0.11 at $K = 8$. Commitments supported
 267 primarily by an agent’s own evidence complete at 0.26–0.34, whereas warp commitments remain
 268 below 0.05. The pattern is stable over provenance thresholds: for $\varphi \in \{0.5, \dots, 0.95\}$, the foreign-
 269 evidence stratum completes at no more than 0.03, while the own-evidence stratum remains between
 270 0.26 and 0.50. This analysis locates the loss in peer-supported commitments, but by itself it does not
 271 distinguish poor evidence from target contention.

272 The reservation-and-rollback ablation separates these explanations. Independent memory removes
 273 peer evidence but leaves physical competition for targets. Reservation retains peer evidence while
 274 retracting conflicting commitments. Under reservation, unconditional success at fast cadence reaches
 275 0.614–0.620, exceeding the $K = 8$ peer baseline of 0.583. Across the fast-cadence cells, the paired
 276 improvement over the same cells without reservation is $\Delta = +0.037$, with cluster-bootstrap 95%
 277 CI $[+0.018, +0.058]$ over the 18 matched design cells ($+0.028, +0.042, +0.043$ at $K = 1, 2, 4$).
 278 The predicted improvement after changing coordination while retaining shared evidence provides
 279 intervention-based evidence that the Completion loss is governed by occupancy contention rather
 280 than by the quality of peer observations. The same directional provenance pattern appears in a
 281 continuous VMAS bridge (0.72 $\rightarrow$ 0.00 across the corresponding strata) and on an LLM text
 282 substrate (0.42 $\rightarrow$ 0.08); full protocols and uncertainty estimates are reported in the supplement.

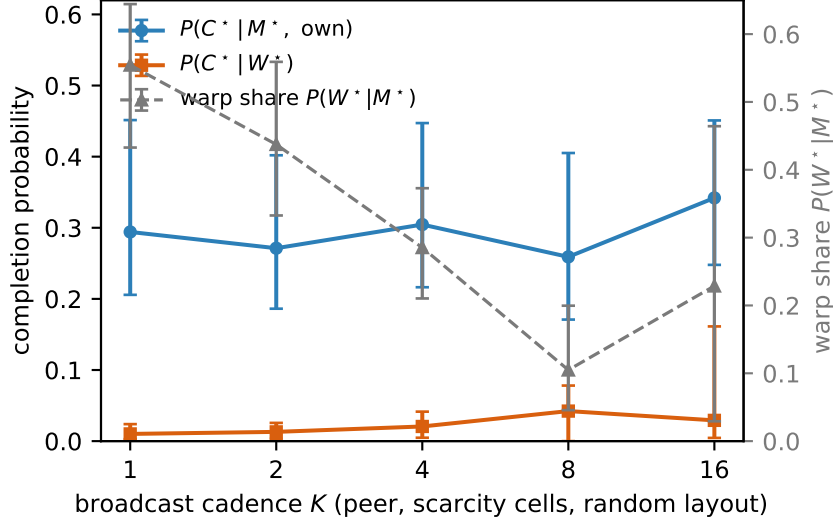


Figure 4: Completion after Materialization stratified by evidence provenance. Faster sharing increases peer-supported locks, whose completion stays below the own-evidence stratum.

283 The robustness checks clarify the scope of stage observability. CommNet-PPO reaches success 0.667
 284 compared with 0.65 for the symbolic peer policy, but its real-valued communication channel does
 285 not expose an informatively empty retrieval event or an explicit target lock, so R and M cannot be
 286 separated without adding probes. In the LLM collective, uncoordinated sharing lowers success from
 287 2/3 to 1/3, reservation recovers part of the loss from 0.33 to 0.42, and a two-model step-budget
 288 sweep maps a 1.00 $\rightarrow$ 0.00 end-to-end collapse to a pure C-stage failure. These checks support the
 289 diagnostic interpretation without constituting a general performance comparison between symbolic
 290 and learned agents.

291 5.4 Portability to third-party agent frameworks

292 The controlled structural faults generate the predicted marginal stage profiles across OpenAI SDK,
 293 LangGraph, and AutoGen. Removing the consolidated memory fact produces $f(R^*) = 0.00$ in 20/20
 294 episodes per framework. Setting the budget below the minimum execution requirement produces
 295 $f(C^*) = 0.00$ while $f(Q^*) = f(R^*) = 1.00$, again in 20/20 episodes per framework. These results
 296 verify the trace adapter and the semantic meaning of the R and C events across the three runtimes
 297 (the full marginal-frequency table and per-variant stage profiles are in the supplement).

298 The clean controls reveal behavior that is not visible from the structural tests. End-to-end completion
 299 ranges from 0.50 to 0.75, first-commitment accuracy is only 0.25–0.50, and re-querying raises
 300 eventual correct commitment to 0.70–0.85. Under stale ambiguity, OpenAI SDK completion falls
 301 from 0.75 to 0.30, whereas the other runtimes recover more often after the initial detour. Repeating all
 302 240 episodes with qwen3:4b preserves the structural R and C localizations but changes commitment
 303 behavior: OpenAI SDK and LangGraph reach 0.90–0.95 first-commitment accuracy, while AutoGen
 304 remains at $f(M_1) = 0.40$. Event semantics are therefore portable, but model–framework behavior is
 305 not interchangeable.

306 The blinded fault study provides the strongest test of diagnostic use in this block. The fixed first-
 307 failing-stage rule localizes 26/30 scored cells for llama3.1 and 29/33 for qwen3:4b; all five structural
 308 faults are localized in all 30/30 model–framework cells, and held-out controls are classified as “none”
 309 in 5/6 cells. Most errors occur at Materialization, where llama3.1’s weak control profile creates a
 310 floor effect that masks additional M-stage perturbations.

311 Compared with baselines (Table 1), Q/R/M/C reaches 0.873 exact-stage accuracy and 0.835 macro-F1,
 312 with Holm-adjusted McNemar comparisons remaining significant against all exact-stage alternatives
 313 (largest adjusted $p = 0.031$). The two-stage R/E baseline matches Q/R/M/C only when scored at
 314 side level; at stage level it loses the commitment/execution distinction. The conformance rule misses

Table 1: Diagnostic baselines on the blinded fault benchmark. 63 scored cells; FPR is the held-out-control false-positive rate, and p is the Holm-adjusted exact McNemar test vs. Q/R/M/C.

Method	Acc.	Macro-F1	FPR	p
Success-only	.095	.077	.00	<.0001
Progress milestones	.730	.698	.00	.023
Two-stage R/E (exact)	.651	.450	.00	.0016
two-stage, side-level	.873	—	—	—
Conformance, first deviation	.778	.755	.17	.031
Learned LR, unseen faults	.444	.435	1.00	<.0001
Q/R/M/C	.873	.835	.17	—

the budget-exhaustion branch, and the learned baseline transfers poorly to unseen fault types while requiring labeled faults.

Repair utility gives the same ordering at coarser resolution. Q/R/M/C attains the highest mean gain and lowest regret ($G = 0.293$, $R = 0.061$, oracle-best gain 0.354), followed by the two-stage baseline (0.280/0.074), conformance (0.250/0.104), and milestone diagnosis (0.219/0.135). Its gain advantage over the milestone baseline is resolved by paired bootstrap ($\Delta G = +0.074$, 95% CI $[+0.007, +0.157]$), while the contrast with the two-stage baseline is not ($+0.013 [-0.009, +0.039]$).

5.5 Limitations and interpretation

The results support Q/R/M/C as a protocol for explicit memory pipelines, not as a universal causal model of agent behavior. The primary substrates are grid worlds; VMAS provides only a directional continuous-control check; and the LLM studies use small local models with modest sample sizes. The protocol assumes an entity correspondence between memory records and world referents. This correspondence is supplied by the instrumented environments and a shared coordinate frame, but remains an open problem for heterogeneous agents in open worlds.

The multi-agent claim is empirical and slope-level. The conditional product is monotone in the observed data, and the $K = 8$ optimum appears only on time-to-success conditioned on success. A censoring-aware metric gives a different ordering. The protocol also operates at one resolution: it identifies Retrieval as the failed link, but separating accumulation, ranking, grounding, or staleness handling requires additional within-link logs. These limits are substantive rather than presentational, and they define the systems and questions for which the reported rates are interpretable.

Taken together, the experiments support a narrower but actionable conclusion. In the single-agent setting, the protocol redirects attention from candidate ranking to memory accumulation. In the multi-agent setting, it separates the benefit of faster evidence consolidation from the cost of conflicting commitments and predicts a coordination intervention that improves unconditional success. In third-party stacks, it preserves structural event semantics while exposing model- and framework-dependent commitment behavior. Q/R/M/C therefore contributes an operational, intervention-validated diagnostic contract: it distinguishes retrieval, commitment, and post-commitment failures, outperforms simpler exact-stage diagnostics, selects repairs that match the strongest coarse alternative without requiring labeled fault training, and explicitly reports when the required stages are not observable.

References

- Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *NeurIPS*, 2017.
- Stephen Boyd, Arpita Ghosh, Balaji Prabhakar, and Devavrat Shah. Randomized gossip algorithms. *IEEE Transactions on Information Theory*, 52(6):2508–2530, 2006.
- Josep Carmona, Boudewijn van Dongen, Andreas Solti, and Matthias Weidlich. *Conformance Checking: Relating Processes and Models*. Springer, 2018.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAs: Automated evaluation of retrieval augmented generation. In *EACL System Demonstrations*, 2024.

354 Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson.
355 Counterfactual multi-agent policy gradients. In *AAAI*, 2018.

356 Saurabh Gupta, James Davidson, Sergey Levine, Rahul Sukthankar, and Jitendra Malik. Cognitive
357 mapping and planning for visual navigation. In *CVPR*, 2017.

358 Martin Leucker and Christian Schallhart. A brief account of runtime verification. *Journal of Logic
359 and Algebraic Programming*, 78(5):293–303, 2009.

360 Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan
361 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *ICLR*, 2024.

362 Xiaofeng Lin, Yingxu Wang, Tung Sum Thomas Kwok, Daniel Guo, Sahil Arun Nale, Charles
363 Fleming, and Guang Cheng. REFLECT: Intervention-supported error attribution for silent failures
364 in LLM agent traces. *arXiv preprint arXiv:2606.09071*, 2026.

365 Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan,
366 Lingpeng Kong, and Junxian He. AgentBoard: An analytical evaluation board of multi-turn LLM
367 agents. In *NeurIPS*, 2024.

368 Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez.
369 MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

370 Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S.
371 Bernstein. Generative agents: Interactive simulacra of human behavior. In *UIST*, 2023.

372 Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, et al. RAGChecker:
373 A fine-grained framework for diagnosing retrieval-augmented generation. In *NeurIPS Datasets
374 and Benchmarks*, 2024.

375 Nikolay Savinov, Alexey Dosovitskiy, and Vladlen Koltun. Semi-parametric topological memory for
376 navigation. In *ICLR*, 2018.

377 Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion:
378 Language agents with verbal reinforcement learning. In *NeurIPS*, 2023.

379 Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with
380 backpropagation. In *NeurIPS*, 2016.

381 Edward C. Tolman. Cognitive maps in rats and men. *Psychological Review*, 55(4):189–208, 1948.

382 Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2016.

383 Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan,
384 and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models.
385 *Transactions on Machine Learning Research*, 2024.

386 Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun
387 Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and
388 Chi Wang. Autogen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv
389 preprint arXiv:2308.08155*, 2023.

390 **NeurIPS Paper Checklist**

391 **1. Claims**

392 Question: Do the main claims made in the abstract and introduction accurately reflect the
393 paper’s contributions and scope?

394 Answer: [\[Yes\]](#).

395 Justification: The abstract, Introduction, and Section 3 state the methodological and empiri-
396 cal claims, including the restriction to observable query–lock chains and the marginal-only
397 treatment of non-nested tool traces.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The Results and Discussion section includes a limitations discussion covering grid-world scope, small LLM studies, entity correspondence assumptions, metric dependence, and within-link resolution limits.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A].

Justification: The paper does not claim novel theoretical results. The conditional product is the ordinary chain rule under nested logged events, and the additional stage-Markov assumption is used only for interpretation of link rates.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.

- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: Section 4 describes the event definitions, interventions, experimental cells, run counts, baselines, repair protocol, and uncertainty procedures needed to reproduce the main claims.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes].

Justification: The submission is prepared to be accompanied by an anonymized code supplement with experiment and analysis scripts; generated environments are specified by code and deterministic seeds rather than by private datasets.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: Section 4 specifies the symbolic, multi-agent, third-party-stack, and repair settings, including run counts, fault cells, baselines, intervention definitions, and the fixed diagnostic thresholds.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: The results report bootstrap confidence intervals, cluster-robust intervals, sign tests, McNemar tests with Holm correction, and paired bootstrap comparisons where these support the empirical claims.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)

- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes].

Justification: Compute resources and software versions are intended to be supplied in the supplemental reproducibility material, consistent with the original reproducibility checklist for this paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes].

Justification: The experiments use synthetic or benchmark environments, tool traces, and local model runs, with no human-subject data collection or deployment decisions described in the paper.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A].

Justification: The work is a diagnostic protocol evaluated in synthetic and benchmark settings and does not introduce a deployed decision system, human dataset, or high-risk model release with direct societal impact claims.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.

- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release pretrained generative models, scraped datasets, or other assets with the high-risk misuse profile contemplated by this question.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No].

Justification: The paper cites existing environments, frameworks, and model/tool stacks, but the current main text does not explicitly enumerate the license and terms of use for each asset. This should be audited before submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes].

Justification: The paper introduces event adapters, benchmark fault specifications, and experiment scripts as research assets, documented through the method description and accompanying code supplement rather than through human-data documentation.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A].

Justification: The paper does not involve crowdsourcing or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A].

Justification: The paper does not involve crowdsourcing or research with human subjects, so IRB or equivalent approval is not applicable.

Guidelines:

- 712 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
713 with human subjects.
- 714 • Depending on the country in which research is conducted, IRB approval (or equivalent)
715 may be required for any human subjects research. If you obtained IRB approval, you
716 should clearly state this in the paper.
- 717 • We recognize that the procedures for this may vary significantly between institutions
718 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
719 guidelines for their institution.
- 720 • For initial submissions, do not include any information that would break anonymity (if
721 applicable), such as the institution conducting the review.

722 16. Declaration of LLM usage

723 Question: Does the paper describe the usage of LLMs if it is an important, original, or
724 non-standard component of the core methods in this research? Note that if the LLM is used
725 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
726 scientific rigor, or originality of the research, declaration is not required.

727 Answer: [Yes].

728 Justification: LLM-based agents are used as experimental substrates in the collective and
729 third-party-stack studies, and the paper reports the model families, tool interfaces, and
730 budget manipulations relevant to those experiments.

731 Guidelines:

- 732 • The answer [N/A] means that the core method development in this research does not
733 involve LLMs as any important, original, or non-standard components.
- 734 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
735 be described.